CHAPTER 7

Symmetric Matrices and Quadratic Forms

(对称矩阵与二次型)

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§ 7.1 Diagonalization of Symmetric Matrices

What is Symmetric Matrices?

- A symmetric matrix is a matrix A such that $A^T = A$.

Example ：

Symmetric:

$$\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & 8 \\ 0 & 8 & -7 \end{bmatrix}, \quad \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

Nonsymmetric:

$$\begin{bmatrix} 1 & -3 \\ 3 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & -4 & 0 \\ -6 & 1 & -4 \\ 0 & -6 & 1 \end{bmatrix}, \quad \begin{bmatrix} 5 & 4 & 3 & 2 \\ 4 & 3 & 2 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}$$

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§ 7.1 Diagonalization of Symmetric Matrices

Properties of Symmetric Matrices

- Such a matrix is necessarily square.
- Its main diagonal entries are arbitrary.
- Its other entries occur in pairs- on opposite sides of the main diagonal.

$$\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & 8 \\ 0 & 8 & -7 \end{bmatrix}, \quad \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}$$

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Review ： the diagonalization process.

Example ： If possible, diagonalize the matrix

$$A=\begin{bmatrix} 6 & -2 & -1 \\ -2 & 6 & -1 \\ -1 & -1 & 5 \end{bmatrix}.$$

Solution： The characteristic equation of A is

$$0=-\lambda^3+17\lambda^2-90\lambda+144=-(\lambda-8)(\lambda-6)(\lambda-3)$$

Standard calculations produce a basis for each eigenspace:

$$\lambda=8:\boldsymbol{v}_1=\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix};\quad \lambda=6:\boldsymbol{v}_2=\begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix};\quad \lambda=3:\boldsymbol{v}_3=\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

- Normailize v1, v2 and v3 to produce the unit eigenvectors

$$\boldsymbol{u}_1 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \quad \boldsymbol{u}_2 = \begin{bmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}, \quad \boldsymbol{u}_3 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Let

$$P = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{bmatrix}, \quad D = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Then $A = PDP^{-1}$. Since P is square and has orthonormal columns, P is an orthogonal matrix, and P^{-1} is simply P^T.

§ 7.1 Diagonalization of Symmetric Matrices

Theorem 1

If A is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.

Proof： Let v_1 and v_2 be eigenvectors that correspond to distinct eigenvalues, say λ_1 and λ_2. To show that $v_1 \cdot v_2 = 0$, compute

$$\begin{aligned}\lambda_1 v_1 \cdot v_2 &= (\lambda_1 v_1)^T v_2 = (Av_1)^T \cdot v_2 \\ &= (v_1^T A^T) v_2 = v_1^T (Av_2) \\ &= v_1^T (\lambda_2 v_2) = \lambda_2 v_1^T v_2 = \lambda_2 v_1 \cdot v_2\end{aligned}$$

$$\therefore \quad (\lambda_1 - \lambda_2) v_1 \cdot v_2 = 0 ,$$

$$\because \quad \lambda_1 \neq \lambda_2 , \qquad \therefore \ v_1 \cdot v_2 = 0 .$$

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- A matrix A is said to be orthogonally diagonalizable if there are an orthogonal matrix（正交矩阵）P (with $P^{-1} = P^T$) and a diagonal matrix D such that

$$A = PDP^T = PDP^{-1} \quad (1)$$

- To orthogonally diagonalize an n×n matrix, we must find n linearly independent and orthonormal eigenvectors.
- If A is orthogonally diagonalizable as in (1), then

$$A^T = (PDP^T)^T = P^{TT} D^T P^T = PDP^T = A$$

§7.1 Diagonalization of Symmetric Matrices

Theorem 2

An n×n matrix A is orthogonally diagonalizable if and only if A is symmetric matrix.

已经说明可正交对角化一定是对称矩阵，

因此只需证明对称矩阵可正交对角化（ 后面内容 ）

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- Example： Orthogonally diagonalize the matrix

$$A=\begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

Its characteristic equation is

$$0=-\lambda^3+12\lambda^2-21\lambda-98=-(\lambda-7)^2(\lambda+2)$$

- Solution： Compute to produce bases for the eigenspaces:

$$\lambda=7:\boldsymbol{v}_1=\begin{bmatrix}1\\0\\1\end{bmatrix},\ \boldsymbol{v}_2=\begin{bmatrix}-1/2\\1\\0\end{bmatrix},\ \lambda=-2:\boldsymbol{v}_3=\begin{bmatrix}-1\\-1/2\\1\end{bmatrix}$$

Although v_1 and v_2 are linearly independent, they are not orthogonal.

- Recall from Section 6.2 that the projection of v_2 onto v_1 is $\frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1$, and the component of v_2 orthogonal to v_1 is

$$z_2 = v_2 - \frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix} - \frac{-1/2}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/4 \\ 1 \\ 1/4 \end{bmatrix}$$

Then $\{v_1, z_2\}$ is an orthogonal set in the eigenspace for $\lambda = 7$.

Since the eigenspace is two-dimensional (with basis v_1, v_2), the orthogonal set $\{v_1, z_2\}$ is an orthogonal basis for the eigenspace.

利用正交化过程对$\mathbf{v_1}$, $\mathbf{v_2}$ 进行正交化

Normalizing v1 and z2, we obtain the following orthonormal basis for the eigenspace for λ = 7:

$$\boldsymbol{u}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, \quad \boldsymbol{u}_2 = \begin{bmatrix} -1/\sqrt{18} \\ 4/\sqrt{18} \\ 1/\sqrt{18} \end{bmatrix}$$

An orthonormal basis for the eigenspace for λ = -2 is:

$$\boldsymbol{u}_3 = \frac{1}{\|2\boldsymbol{v}_3\|} \cdot 2\boldsymbol{v}_3 = \frac{1}{3}\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}$$

By Theorem 1, u_3 is orthogonal to the other eigenvectors u_1 and u_2. Hence $\{u_1,u_2,u_3\}$ is an orthonormal set.

Let

$$P=[\boldsymbol{u}_1\ \boldsymbol{u}_2\ \boldsymbol{u}_3]=\begin{bmatrix}1/\sqrt{2} & -1/\sqrt{18} & -2/3\\ 0 & 4/\sqrt{18} & -1/3\\ 1/\sqrt{2} & 1/\sqrt{18} & 2/3\end{bmatrix},D=\begin{bmatrix}7 & 0 & 0\\ 0 & 7 & 0\\ 0 & 0 & -2\end{bmatrix}$$

Then P orthogonally diagonalizes A, and $A = PDP^{-1}$.

§7.1 Diagonalization of Symmetric Matrices

The Spectral Theorem（ 谱定理 ）

Theorem 3 The Spectral Theorem for Symmetric Matrices

An n×n symmetric matrix A has the following properties:

a. A has n real eigenvalues, counting multiplicities.(p341:23,24)

b. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.

c. The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.

d. A is orthogonally diagonalizable.

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§ 7.1 Diagonalization of Symmetric Matrices

Spectral Decomposition

Suppose $A = PDP^{-1}$, where the columns of P are orthonormal eigenvectors $u_1,\ldots,u_n$ of A and the corresponding eigenvalues $\lambda_1,\ldots,\lambda_n$ are the diagonal matrix D. Then, since $P^{-1} = P^T$,

$$A = PDP^T = [u_1 \cdots u_n]\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}\begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$

$$= [\lambda_1 u_1 \cdots \lambda_n u_n]\begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$

- Using the column-row expansion of a product, we can write

$$A = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \cdots + \lambda_n u_n u_n^T$$

This representation of A is called **a spectral decomposition**（ 谱分解 ）of A because it breaks up A into piece determined by the spectrum (eigenvalues) of A.

- Example ： Construct a spectral decomposition of the matrix A that has the orthogonal diagonalization

$$A=\begin{bmatrix}7 & 2\\ 2 & 4\end{bmatrix}=\begin{bmatrix}2/\sqrt{5} & -1/\sqrt{5}\\ 1/\sqrt{5} & 2/\sqrt{5}\end{bmatrix}\begin{bmatrix}8 & 0\\ 0 & 3\end{bmatrix}\begin{bmatrix}2/\sqrt{5} & 1/\sqrt{5}\\ -1/\sqrt{5} & 2/\sqrt{5}\end{bmatrix}$$

Solution： Denote the columns of P by u_1 and u_2. Then

$$A=8\boldsymbol{u}_1\boldsymbol{u}_1^{\mathrm{T}}+3\boldsymbol{u}_2\boldsymbol{u}_2^{\mathrm{T}}$$

To verify this decomposition of A, compute

$$\boldsymbol{u}_1\boldsymbol{u}_1^{\mathrm{T}}=\begin{bmatrix}2/\sqrt{5}\\ 1/\sqrt{5}\end{bmatrix}\begin{bmatrix}2/\sqrt{5} & 1/\sqrt{5}\end{bmatrix}=\begin{bmatrix}4/5 & 2/5\\ 2/5 & 1/5\end{bmatrix}$$

$$\boldsymbol{u}_2\boldsymbol{u}_2^{\mathrm{T}}=\begin{bmatrix}-1/\sqrt{5}\\ 2/\sqrt{5}\end{bmatrix}\begin{bmatrix}-1/\sqrt{5} & 2/\sqrt{5}\end{bmatrix}=\begin{bmatrix}1/5 & -2/5\\ -2/5 & 4/5\end{bmatrix}$$

$$8\boldsymbol{u}_1\boldsymbol{u}_1^{\mathrm{T}}+3\boldsymbol{u}_2\boldsymbol{u}_2^{\mathrm{T}}=\begin{bmatrix}32/5 & 16/5\\ 16/5 & 8/5\end{bmatrix}+\begin{bmatrix}3/5 & -6/5\\ -6/5 & 12/5\end{bmatrix}=\begin{bmatrix}7 & 2\\ 2 & 4\end{bmatrix}=A$$

Assignment

- P431：17,27

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§7.2 Quadratic Forms（二次型）

- 1. Definition of Quadratic Forms
- 2. Change of Variable in a Quadratic Form
- 3. A Geometric View of Principal Axes
- 4. Classifying Quadratic Forms

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§ 7.2 Quadratic Forms

1. Definition of Quadratic Forms

A quadratic form on R^n is a function Q defined on R^n whose value at a vector x in R^n can be computed by an expression of the form $Q(x) = x^TAx$, where A is an nxn symmetric matrix.

The matrix A is called the matrix of the quadratic form（二次型矩阵）.

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- example ： Let $x=\begin{bmatrix}x_1\\x_2\end{bmatrix}$, Compute x^TAx for the following matrices:

$$\text{a.}\quad A=\begin{bmatrix}4&0\\0&3\end{bmatrix} \qquad \text{b.}\quad A=\begin{bmatrix}3&-2\\-2&7\end{bmatrix}$$

- Solution：

$$\text{a.}\quad x^TAx=[x_1\ \ x_2]\begin{bmatrix}4&0\\0&3\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=[x_1\ \ x_2]\begin{bmatrix}4x_1\\3x_2\end{bmatrix}=4x_1^2+3x_2^2.$$

b. There are two -2 entries in A.

$$\begin{aligned}x^TAx&=[x_1\ \ x_2]\begin{bmatrix}3&-2\\-2&7\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}=[x_1\ \ x_2]\begin{bmatrix}3x_1-2x_2\\-2x_1+7x_2\end{bmatrix}\\&=x_1(3x_1-2x_2)+x_2(-2x_1+7x_2)\\&=3x_1^2-2x_1x_2-2x_2x_1+7x_2^2\\&=3x_1^2-4x_1x_2+7x_2^2\end{aligned}$$

- Example ： For x in R^3, let $Q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$, write this quadratic form as x^TAx.

Solution： The coefficient of x_1^2, x_2^2, x_3^2 go on the diagonal of A.

To make A symmetric, the coefficient of x_ix_j for $i \neq j$ must be split evenly between the (i, j)- and (j, i)–entries in A. The coefficient of x_1x_3 is 0.

$$Q(x) = x^TAx = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

§ 7.2 Quadratic Forms

2. Change of Variable in a Quadratic Form

If x represents a variable vector in R^n, then a change of variable is an equation of the form

$x = Py$ or $y = P^{-1}x$（变量代换）

where P is an invertible matrix and y is a new variable vector in R^n.

If the change of variable is made, then

$$x^T Ax = (Py)^T A(Py) = y^T P^T APy = y^T (P^T AP)y$$

If P orthogonally diagonalizes A, then…..

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- Example ： Make a change of variable that transforms the quadratic form into a quadratic form with no cross-product term.

$$Q(\boldsymbol{x}) = x_1^{\,2} - 8x_1x_2 - 5x_2^2 ,$$

- Solution： The matrix of the quadratic form is

$$A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$$

Orthogonally diagonalize A:

$$\lambda = 3 : \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix}; \qquad \lambda = -7 : \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

- These vectors are orthogomal and so provide an orthonormal basis for R^2. Let

$$P = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 0 \\ 0 & -7 \end{bmatrix}$$

Then $A = PDP^{-1}$ and $D = P^{-1}AP = P^{T}AP$.

A suitable change of variable is

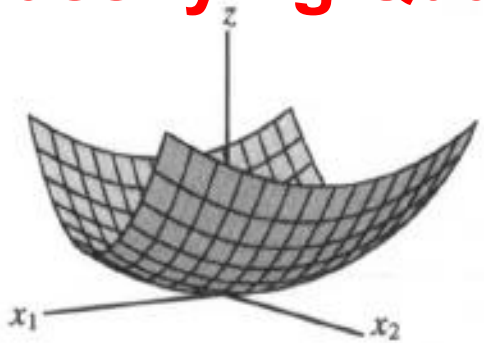
$$\boldsymbol{x} = P\boldsymbol{y}, \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Then

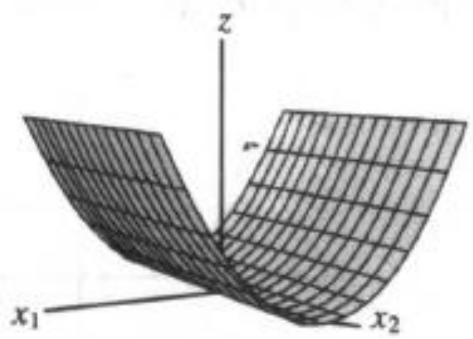
$$\begin{aligned} x_1^2 - 8x_1x_2 - 5x_2^2 &= \boldsymbol{x}^T A\boldsymbol{x} \\ &= (P\boldsymbol{y})^T A(P\boldsymbol{y}) \\ &= \boldsymbol{y}^T P^T AP\boldsymbol{y} = \boldsymbol{y}^T D\boldsymbol{y} \\ &= 3y_1^2 - 7y_2^2 \end{aligned}$$

§ 7.2 Quadratic Forms

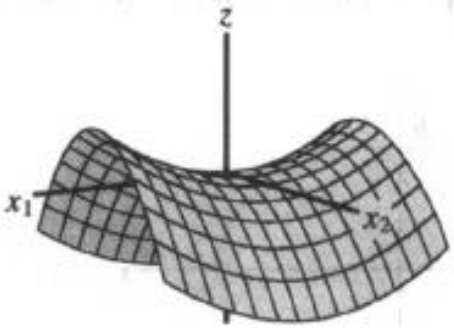
3. Classifying Quadratic Forms



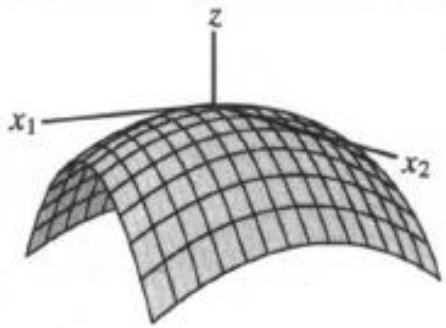
a） $z = 3x_1^2 + 7x_2^2$



b） $z = 3x_1^2$



c） $z = 3x_1^2 - 7x_2^2$



d） $z = -3x_1^2 - 7x_2^2$

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§7.2 Quadratic Forms

- Definition：

A quadratic form Q is:

a. positive definite（ 正定的 ）, if $Q(x) > 0$ for all $x \neq 0$.

b. negative definite（ 负定的 ）, if $Q(x) < 0$ for all $x \neq 0$.

c. indefinite（ 不定的 ）, if Q(x) assumes both positive and negative values.

positive semidefinite(半正定的)

negative semidefinite（ 半负定的 ）

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§ 7.2 Quadratic Forms

Theorem 5 Quadratic Forms and Eigenvalues

Let A be an n×n symmetric matrix. Then a quadratic form x^TAx is:

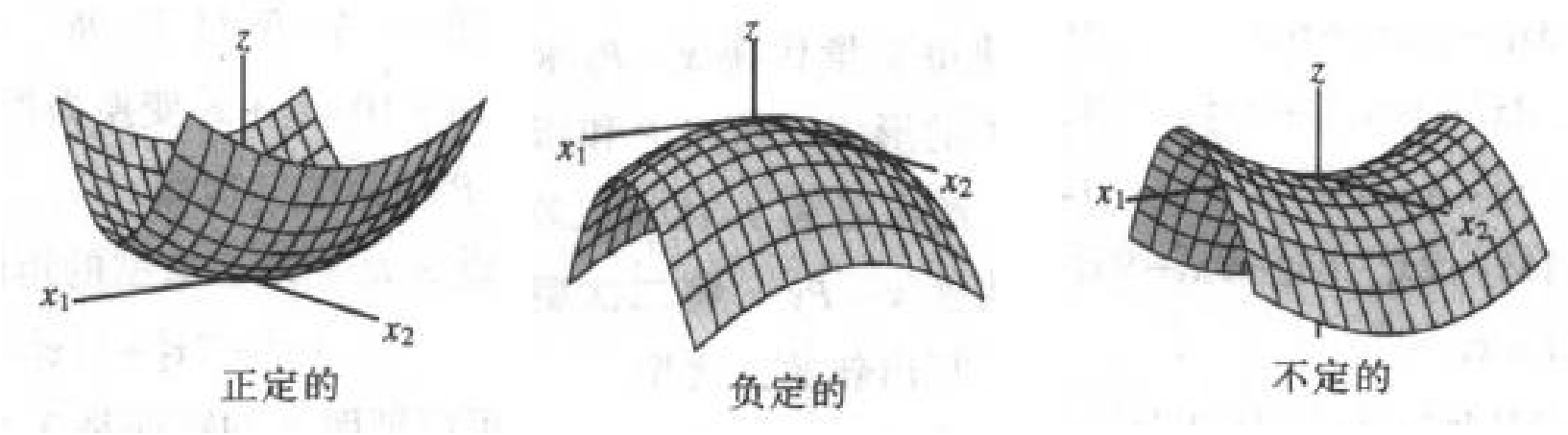
a. positive definite if and only if the eigenvalues of A are all positive.

b. negative definite if and only if the eigenvalues of A are all negative

c. indefinite if and only if A has both positive and negative eigenvalues.

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§ 7.2 Quadratic Forms



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Assignment

- P438:7,23,24

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